(01) $(0,25)^{100} \cdot 2^{2001} = \left(\frac{1}{4}\right)^{1000} \cdot 2^{2001} = \left(\frac{1}{2^2}\right)^{1000} \cdot 2^{2001}$

$= \frac{1}{2^{2000}} \cdot 2^{2001} = 2^{2001-2000} = \underline{\underline{2}}$

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(02) $V_m = \frac{\Delta s}{\Delta t} \Leftrightarrow 3 \cdot 10^5 = \frac{150 \cdot 10^6}{\Delta t} \Leftrightarrow \Delta t = 500\ s$

logo $\Delta t = \frac{500}{60} \cong 8,33$ MIN.

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(03) $\frac{(5,2)^4 \cdot (10,3)^3}{(9,9)^2} = \frac{5^4 \cdot 10^3}{10^2} = 5^4 \cdot 10 = 6250$

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(04) $\frac{2^{10} - 3^6}{2^5 + 3^3} = \frac{1024 - 729}{32 + 27} = \frac{295}{59} = \underline{\underline{5}}$

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(05) SENDO h a altura da MENINA:

$25 = \frac{64}{h^2} \Leftrightarrow h^2 = \frac{64}{25} \Leftrightarrow h = \frac{8}{5} \Leftrightarrow h = 160$ cm

logo: $RIP = \frac{160}{\sqrt[3]{64}} = \frac{160}{4} = 40$

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(06) $\sqrt{13 + \sqrt{7 + \sqrt{2 + \sqrt{4}}}} = \sqrt{13 + \sqrt{7 + \sqrt{2 + 2}}}$

$= \sqrt{13 + \sqrt{7 + \sqrt{4}}} = \sqrt{13 + \sqrt{7 + 2}} = \sqrt{13 + \sqrt{9}} = \sqrt{13 + 3}$

$= \sqrt{16} = \underline{\underline{4}}$

07

$x^3 = 25$

$x^3 = 5^2$

$\sqrt{x^3} = 5$

$\boxed{x^{3/2} = 5}$

$y^2 = 27$

$y^2 = 3^3$

$y = \sqrt{3^3}$

$\boxed{y = 3^{3/2}}$

$x^{3/2} \cdot y^{4/3} = 5 \cdot \left(3^{3/2}\right)^{4/3} = 5 \cdot 3^2 = \underline{\underline{45}}$

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08 $(256)^{-3/4} = \left(2^8\right)^{-3/4} = 2^{-6} = \frac{1}{2^6} = \underline{\underline{\frac{1}{64}}}$

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09 $P(t) = k \cdot 2^{0,05 \cdot t}$

- $P(0) = 300 \Leftrightarrow k \cdot 2^{0,05 \cdot 0} = 300 \Leftrightarrow \underline{k = 300}$
- $P(10) = 300 \cdot 2^{0,05 \cdot 10} \Leftrightarrow P(10) = 300 \cdot 2^{0,5} \Leftrightarrow$

$P(10) = 300 \cdot \sqrt{2} \Leftrightarrow P(10) = 300 \cdot 1,41 \Leftrightarrow$

$P(10) \cong 423$ mil habitantes

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10 $a = 3^3 \Leftrightarrow a = 27$

a) $b = (-2)^3 \Leftrightarrow b = -8$

$c = 3^{-2} \Leftrightarrow c = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$

$d = (-2)^{-3} \Leftrightarrow d = \left(-\frac{1}{2}\right)^3 \Leftrightarrow d = -\frac{1}{8}$

b) $-8, -\frac{1}{8}, \frac{1}{9}, 27$ ou $b < d < c < a$

(11) $S(p) = \frac{11}{100} \cdot p^{2/3}$

a) $S(8) = \frac{11}{100} \cdot 8^{2/3} \Leftrightarrow S(8) = \frac{11}{100} \cdot \sqrt[3]{8^2}$

$\Leftrightarrow S(8) = \frac{11}{100} \cdot 4 \Leftrightarrow \underline{S(8) = 0{,}44\ m^2}$

b) $2 \cdot 0{,}44 = \frac{11}{100} \cdot p^{2/3} \Leftrightarrow 0{,}88 = \frac{11}{100} \cdot \sqrt[3]{p^2} \Leftrightarrow$

$\frac{88}{11} = \sqrt[3]{p^2} \Leftrightarrow 8 = \sqrt[3]{p^2} \Leftrightarrow 8^3 = \left(\sqrt[3]{p^2}\right)^3$

$\Leftrightarrow 512 = p^2 \Leftrightarrow p = \sqrt{512} \Leftrightarrow p = \sqrt{2^2 \cdot 2^2 \cdot 2^2 \cdot 2^2 \cdot 2}$

$\Leftrightarrow p = 16 \cdot \sqrt{2} \Leftrightarrow \underline{p \cong 22{,}6\ kg}$

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(12) c) $(a^x)^y = a^{x \cdot y}$ e NÃO a^{xy}.

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(13) NA LISTA!

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(14) NA LISTA!

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(15) $\frac{4 \cdot 2^x}{4^x} = \frac{1}{64} \Leftrightarrow \frac{2^2 \cdot 2^x}{2^{2x}} = \frac{1}{2^6} \Leftrightarrow \frac{2^{2+x}}{2^{2x}} = 2^{-6}$

$\Rightarrow 2^{2+x-2x} = 2^{-6} \Leftrightarrow 2^{2-x} = 2^{-6} \Leftrightarrow 2 - x = -6$

$\Leftrightarrow x = 8$

(16) I) FALSA, pois $-3^2 = -9$

II) FALSA, pois $a^2 + 2ab + b^2$

III, IV, V $\Rightarrow$ verdadeiras

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(18) $2^{100} \cdot 5^{103} = 2^{100} \cdot 5^{100} \cdot 5^3 \Leftrightarrow (2 \cdot 5)^{100} \cdot 5^3$

$= 10^{100} \cdot 5^3 = 125 \cdot 10^{10} = 125.\underbrace{000.000 \ldots 000}_{100 \text{ zeros}}$

logo a soma dos algarismos é $1+2+5+0+0+\ldots+0 = 8$

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(19) Na lista!

(17) a) Substituindo $B = 600$ KCAL e $M = 16$ kg, temos: $B = K \cdot M^{3/4}$

$$600 = K \cdot 16^{3/4} \Leftrightarrow 600 = K \cdot (2^4)^{3/4}$$

$$\Leftrightarrow 600 = K \cdot 2^3 \Leftrightarrow K = \frac{600}{8} = \underline{\underline{75}}$$

b) $B_G = K \cdot G^{3/4}$ e $B_P = K \cdot P^{3/4}$

Pelo enunciado, $G = 10^6 \cdot P$, logo

$$\frac{B_G}{B_P} = \frac{K \cdot (10^6 \cdot P)^{3/4}}{K \cdot P^{3/4}} = \frac{10^{\frac{18}{4}} \cdot P^{3/4}}{P^{3/4}}$$

$$\frac{B_G}{B_P} = 10^{9/2} = \sqrt{10^9} = \sqrt{10^2 \cdot 10^2 \cdot 10^2 \cdot 10^2 \cdot 10}$$

$$\frac{B_G}{B_P} = 10^4 \cdot \sqrt{10} = 10^4 \cdot \underbrace{3{,}1623}_{\text{TABELA}} = \underline{\underline{31.623}}$$